UDF Applications

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object or variable

class - type

```
def reverseList(li):
    if isinstance(li, list):
        li.reverse()
        return li
    else:
        return f'{li} is not a list'
```

check the data type or (class type) or sub-class type and it returns boolean value.

```
import random as R
li = [ R.randint(10, 99) for _ in range(11)]
print(li)
revLi = reverseList(li)
print(revLi)
revLi = reverseList("Hello!")
print(revLi)
```

Write a function def equals(a, b) that checks whether two lists have the same elements in the same order

```
def equals(a, b):
    if isinstance(a, list) and isinstance(b, list):
        return a == b
    else:
        return f'{a} or {b} is not a list'
```

module - 1

```
def listInput():
    x = input("Enter elements of the list in [1, 2, 3, 'hello']: ")
    li = eval(x)
    return li
```

module - 2

```
# main code
li1 = listInput()
li2 = listInput()
if equals(li1, li2):
    print('Both lists having same elements.')
else:
    print('Both lists have different elements')
```

module - 3.

Write a function def sameSet(a, b) that checks whether two lists have the same ele ments in some order, ignoring duplicates.
For example, the two lists [1 4 9 16 9 7 4 9 11] and [11 11 7 9 16 4 1] would be considered identical. You will probably need one or more helper functions.

```
def listInput():
    x = input("Enter elements of the list in [1, 2, 3, 'hello']: ")
    li = eval(x)
    return li
def sameSet(a:list, b:list):
    if isinstance(a, list) and isinstance(b, list):
        for e in b:
            if e not in a:
                return False  # b is not a subset
        return True # b is a subset
    else:
```

```
        return f'{a} or {b} may not be a list'

li1 = listInput()
li2 = listInput()
isSubSet = sameSet(li1, li2)
if isinstance(isSubSet, bool):
    if isSubSet:
        print('list b is the subset')
    else:
        print('list b is not a subset')
else:
    print(isSubSet)
```

Output:

```
Enter elements of the list in [1, 2, 3, 'hello']: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

Enter elements of the list in [1, 2, 3, 'hello']: [4, 8, 2, 2, 1, 5]
list b is the subset
PS C:\Users\bluej\Documents\Students\JiyaGaurDeensAcademy\Coding> & "C:/Program F
iles/Python312/python.exe" c:/Users/bluej/Documents/Students/JiyaGaurDeensAcademy
/Coding/UDF/Pg374P12.py
Enter elements of the list in [1, 2, 3, 'hello']: [2, 'hello', 'jiya']
Enter elements of the list in [1, 2, 3, 'hello']: [2, 'hello', 'jiya', 'jiya']
list b is the subset
PS C:\Users\bluej\Documents\Students\JiyaGaurDeensAcademy\Coding> & "C:/Program F
iles/Python312/python.exe" c:/Users/bluej/Documents/Students/JiyaGaurDeensAcademy
/Coding/UDF/Pg374P12.py
Enter elements of the list in [1, 2, 3, 'hello']: [2, 3, 4, 5]
Enter elements of the list in [1, 2, 3, 'hello']: [2, 2, 1, 3]
list b is not a subset
PS C:\Users\bluej\Documents\Students\JiyaGaurDeensAcademy\Coding>
```